HW 9

CSE 331

Professor Kevin Zatloukal

Due: **Saturday** December 2nd, 2024 11:00pm

December 7, 2024

Task 1 - He's Making a Dist, and Checking It Twice

(a) Write an expression that is true iff location ℓ falls within the region R.

Solution: $R.x_1 \leq \ell.x \leq R.x_2 \;\; \wedge \;\; R.y_1 \leq \ell.y \leq R.y_2$

(b) Write an expression that is true iff location ℓ is directly below R.

Solution: $R.x_1 \leq \ell.x \leq R.x_2 \;\; \wedge \;\; \ell.y > R.y_2$

(c) Write an expression that calculates distance from ℓ to the closest location in R with ℓ directly below R.

Solution: $\ell.y - R.y_2$

(d) Write an expression that is true iff location ℓ is below and right of R.

Solution: $\ell.x > R.x_2 \;\; \wedge \;\; \ell.y > R.y_2$

(e) Write an expression that calculates the distance from ℓ to the closest location in R when ℓ is below and right of R. Feel free to use the function dist defined before.

Solution: $\mathrm{dist}(\ell, \{x : R.x_2,\, y : R.y_2\})$

Task 2

(a) Prove that, if the region R contains all locations in the tree T, then cl(T), ℓ, R, c) = closest(T, ℓ, c). Your proof should be by structural induction on T.

Solution: Let $P(T)$ be the statement that if the region R contains all locations in the tree T, then cl(T), ℓ, R, c) = closest(T, ℓ, c). We want to prove $P(T)$ for any T by structural induction on T.

- Base Cases:
  1. T = empty, i.e., P(empty)

$$\begin{aligned} \mathrm{cl}(\texttt{empty}, \ell, R, c) &= c && \text{def of cl} \\ &= \mathrm{closest}(\texttt{empty}, \ell, c) && \text{def of closest} \end{aligned}$$

  2. T = single(s), i.e., P(single(s))
  For any single(s), note that, if R does not contain s, then P is vacuously true, so we continue under the assumption that R contains s. We continue by cases.
     - Suppose dist(s, c) $\leq$ dist(s, l) is true. In this case,

$$\begin{aligned} \mathrm{cl}(\mathrm{single}(s), \ell, R.c) &= c && \text{def of cl (since dist}(s,\ c) \leq \text{dist}(s,\ \ell)\text{)} \\ &= \mathrm{closest}(\mathrm{single}(s),\ \ell,\ c)) && \text{def of closest (since dist}(s,\ c) \leq \text{dist}(s,\ \ell)\text{)} \end{aligned}$$

     - Now, suppose dist(s, c) $>$ dist(s, l) is true. In this case,

$$\begin{aligned} \mathrm{cl}(\mathrm{single}(s), \ell, R.c) &= s && \text{def of cl (since dist}(s,\ c) > \text{dist}(s,\ \ell)\text{)} \\ &= \mathrm{closest}(\mathrm{single}(s),\ \ell,\ c)) && \text{def of closest (since dist}(s,\ c) > \text{dist}(s,\ \ell)\text{)} \end{aligned}$$

  These two cases are exhaustive, so we have completed the proof for single(s).

- Inductive Hypothesis:
  Assume that $P(\texttt{nw}), P(\texttt{ne}), P(\texttt{sw}), P(\texttt{se})$ holds for some trees `nw`, `ne`, `se`, `sw`.
- Inductive Steps:
  We need to show $P(S)$ for some tree $S = \texttt{split}(m,\ \texttt{nw},\ \texttt{ne},\ \texttt{sw},\ \texttt{se})$.
  Again, note that, if R does not contain m, then P is vacuously true, so we continue under the assumption that R contains m.
     - Suppose that dist(ℓ, c) $\leq$ distTo(ℓ, R) is true:
     If the distance from ℓ to c is less than or equal to the distance from ℓ to R , then all points in R are farther from ℓ than c and closest will return c by the given fact. Thus:

$$\begin{aligned} \mathrm{closest}(S, \ell, c) &= c && \text{given fact} \\ &= \mathrm{cl}(S, \ell, R, c) && \text{def of cl} \end{aligned}$$

     - Now, suppose that dist(ℓ, c) $>$ distTo(ℓ, R) is true:

$$\begin{aligned} \mathrm{cl}(S, \ell, R, c) &= \mathrm{cl}(\texttt{nw}, \ell, NW(m, R), \\ &\quad \mathrm{cl}(\texttt{sw}, \ell, SW(m, R), \\ &\quad\quad \mathrm{cl}(\texttt{se}, \ell, SE(m, R), \\ &\quad\quad\quad \mathrm{cl}(\texttt{ne}, \ell, NE(m, R), c)))) && \text{def of cl} \\ &= \mathrm{closest}(\texttt{nw}, \ell, \\ &\quad \mathrm{closest}(\texttt{sw}, \ell, \\ &\quad\quad \mathrm{closest}(\texttt{se}, \ell, \\ &\quad\quad\quad \mathrm{closest}(\texttt{ne}, \ell, c)))) && \text{I.H.} \\ &= \mathrm{closest}(S, \ell, c) && \text{def of closest} \end{aligned}$$

These two cases are exhaustive, so we have completed the proof for $P(S)$.

- Conclusion:
  By structural induction on T, we have shown $P(T)$ for all tree T, i.e., $\mathrm{cl}(T, \ell, R, c) = \mathrm{closest}(T, \ell, c)$, provided that R contains all locations in T.

(b) Explain why your proof still work if we defined cl to traverse in c.c.w. order or any other order we choose.

Solution: Each recursive call independently computes the closest point within its quadrant (and in I.H., the four hypotheses are defined independently.) Finally, the result combines the distances to determine the closest point, regardless the order.